Niklas Baier

ML / AI Engineer

Interdisciplinary professional driven to make a meaningful impact in integrous teams.

Contact

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Key Skills

Software Development

- *Python* development, *fastAPI*, *SQLAlchemy*, *numpy* / *pandas*, unit & integration testing, *BDD* / *DDD*
- Vertical slice feature refinement & system / component design
- DevOps with *Docker*, GitLab CI/CD, *Bash* scripts
- Rapid prototyping of UIs & web apps with *streamlit*
- AWS Cloud & IaC with *Terraform*

Machine Learning & GenAI

- solid understanding of traditional ML algorithms
- Prompt engineering & RAG
- Familiarity with *Metaflow* & Databricks-managed *MLflow* for model training & evaluation

Product Management

- Requirements engineering
- Stakeholder management
- Agile methodologies, incl. SCRUM

Soft Skills

- Effective cooperation & strong communication
- Presentation skills

Languages

- German - native
- English - C2 (DAAD)

Experience

Associate Machine Learning Engineer

ProSiebenSat.1 Tech & Services GmbH 01/2024 - present

Tech Lead for company-wide GenAI web app with thousands of internal users

- Designed a zero-downtime process for model deprecation affecting 800+ productively used Assistants (custom GPTs)
- Led strategic direction as interim Product Owner, spearheading vision & roadmap while continuing development contributions
- Engineered a suite of automated smoke & online integration tests to ensure availability of model offering across cloud providers (AWS Bedrock, Azure AI Foundry, Google Vertex AI)
- Conceptualized & implemented a text-to-image feature using OpenAI's DALL-E 3 diffusion model, enabling more than 100k generations

Contributions to department-wide AI Services Platform

- Configured & deployed LibreChat on the AWS Cloud using IaC with Terraform, *AWS EC2*, *ALB*, *Route53*, and *Azure EntraID* SSO integration as a scalable GenAI platform for text & image model inferencing
- Built a retrieval evaluation pipeline using *MLflow* & applied contextual embedding strategy to increase retrieval performance significantly across document- & chunk-level metrics
- Authored a *Python* package to drive reproducible LLMOps prompt engineering experiments via *MLflow*, for stand-alone system prompts and end-to-end evaluation of RAG pipelines

Data Science Intern

ProSiebenSat.1 Tech & Services GmbH 07/2023 - 12/2023

- Spearheaded the development of an interactive *streamlit* web app for machine learning model explainability via *shap*, and ongoing productive model performance monitoring
- Engineered and tested a new competition feature end-to-end via *Metaflow*, *MLflow* and the AWS Cloud for linear TV market share forecasting
- Strengthened *Python* and *SQL* proficiency through exploratory data & detailed error analysis, as well as regular code reviews
- Leveraged the OpenAI API and advanced prompt engineering techniques to augment the project's training dataset with movie franchises
- Involved actively in business stakeholder, data science, and department-wide reviews

Quantitative Research Thesis Student

Allianz Global Investors 07/2022 - 01/2023

- Conducted Master Thesis (graded 1.0) under supervision of the Systematic Equity team, training modern machine & deep learning models with *Python* on historical financial & macroeconomic data to forecast corporate earnings
- Leveraged these forecasts to simulate an equity investment strategy within the Azure Cloud (*Data Lake*, *Databricks*)

Side Projects

Instant Repo

Template repository for rapid creation of new projects from a well-established starting base, using *poetry*, *pre-commit hooks*, a skeleton CI/CD & more

Securities Master Database

Financial data repository deployed on a local Raspberry Pi server, storing and automatically updating data for over 10k financial instruments from the Yahoo Finance & Alpaca Trade APIs

About Me

Interests

- Traditional & Generative AI
- PC Hardware & Mechanical Keyboards
- Virtual Reality

Hobbies

- Strength & Endurance Training
- Electric & Acoustic Guitar
- Video- & Board Games (RPGs)
- Creative (Video) Projects

IT Solution Advisory Working Student

McKinsey & Company 05/2022 - 06/2023

- Demonstrated strong problem-solving skills and effective communication while providing time-critical hardware and software support in an international first-level IT user support role

Quantitative Research Intern

Quantumrock 01/2022 - 04/2022

- Utilized *Python* to develop and backtest a quantitative investment strategy focused on capitalizing on mean reversion effects in multi-factor momentum investing, in close collaboration with the CIO and the Heads of Algorithmic Trading & AI Systems
- Summarized the findings in an academic research paper using *LaTeX*

Other Working Student & Intern Positions

BayernLB (Project & Business Management) 11/2018 - 08/2021

KPMG (Data & Trend Analytics Consulting) 03/2018 - 05/2018

Munich Network (Innovation Sourcing) 10/2016 - 09/2017

Education

M.Sc. Management & Technology

Technical University of Munich 10/2019 - 09/2023

- graduated with final grade of 1.4
- Master Thesis on *Machine Learning for Earnings Forecasting*, graded 1.0
- Major in Finance & Computer Science

B.Sc. Management & Technology

Technical University of Munich 10/2014 - 09/2019

- graduated with final grade of 2.0
- Bachelor Thesis on the *Impact of Meaningful Work on Stock Market Performance*, graded 1.3 & published in *Review of Business Studies* journal
- Major in Business Administration & Computer Science

German Abitur

Municipal High School Werner-von-Siemens 08/2006 - 07/2014

- graduated as Valedictorian with final grade of 1.0

International Studies

Bocconi University (Summer School) 06/2021 - 07/2021

HEC Paris (Summer School) 06/2020 - 07/2020

Tohoku University (Exchange Semester) 10/2017 - 03/2018